Sahaj Bhadja

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Summary

System test engineer with 2+ years building **Python automation, real-time software, simulation, and validation pipelines**; experienced translating requirements into test cases, executing coverage-focused tests, analyzing results, documenting investigations, and deploying reliable systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Developed **Python validation and reconciliation automation** for 5,000+ monthly workflows, testing explicit state transitions, malformed data, integration failures, and recovery behavior while sustaining 99%+ production reliability.
- Created milestone-based test scenarios for concurrent services, measuring throughput, latency, and failure isolation across 1,000+ jobs while improving processing performance 3x and reducing dashboard latency 71%.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Prepared and executed **software validation test cases** across PDF, DOCX, image, and ZIP requirements, reproducing failures and implementing secure input checks, fallback behavior, persistent results, and fault-tolerant execution.
- Wrote Python scripts for multithreaded test automation, processing 1,000+ files per batch and 10,000+ daily requests while maintaining stable peak-load execution and reducing operational turnaround time 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **closed-loop vehicle simulation and evaluation pipelines** across 10,000 autonomous-driving episodes, integrating HD-map geometry, temporal trajectories, and system constraints to improve long-horizon planning 18% and behavioral stability 15%.
- Specified repeatable test conditions, collected quantitative metrics, and investigated off-road and lane-following failures, accurately documenting model behavior, system constraints, experiment results, and recommended configuration changes for researchers.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Configured **cross-platform test environments** on Windows, Linux, and macOS VMs, automating nightly builds, installation, integration testing, coverage summaries, failure logs, and reports while validating 20+ unit and integration tests.
- Collaborated with Microsoft and GitHub engineers to trace failed synchronization flows, isolate API and data defects, implement corrections, and deploy controlled releases that improved response times 30% and pipeline efficiency 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Executed **real-time integration and system tests** across eight Node.js components and 2M+ records, validating authentication, resource allocation, notifications, WebSocket synchronization, and Azure SQL behavior before staged deployment.
- Used Sentry, Azure Monitor, and Kibana to reproduce failures, record test conditions, and report investigation results, enabling cross-functional teams to ship verified Docker releases through Jenkins 50% faster.

Skills

Languages: Python, C++, C, Java, Bash, JavaScript, SQL
Simulation: Closed-loop simulation, system models, real-time pipelines, autonomous-driving evaluation, performance analysis
Validation: Test plans, test cases, test automation, coverage, integration testing, functional testing, regression testing
Systems: Linux, Windows, multithreading, TCP/IP, REST APIs, WebSockets, metrics, failure isolation
Tools: PyTest, Mockito, Selenium, Git, Jenkins, GitHub Actions, Docker, Sentry, Azure Monitor, Kibana

Projects

CPU Scheduling Simulator Live | *C++, Simulation, Validation*

- Implemented and validated **six C++ scheduling algorithms** with interactive controls, Gantt visualization, edge-case testing, and dynamic performance metrics across 1,000+ process combinations, reducing average turnaround time 25%.

Linguistix | *Python, AWS, Docker, CI/CD*

- Developed and deployed a **containerized Python validation service** on AWS ECS/Fargate with CodePipeline, automated quality checks, and repeatable CI/CD releases, achieving 87% accuracy and production-grade operational reliability at scale.

CaptionCraft Live | *Python, PyTorch, Computer Vision*

- Designed a **repeatable model-validation benchmark** across four vision architectures, 10,000 images, and 100,000 captions, analyzing failures and test results to improve accuracy from 81% to 96% while building inference APIs.